1. Equivalence.

(binary quad form) $f(x,y) = ax^2 + bxy + cy^2$

(discriminant) $d = b^2 - 4ac$

(principal form) $\begin{cases} x^2 - \frac{1}{4}dy^2 & d \equiv 0 \pmod 4 \\ x^2 + xy + \frac{1}{4}(1-d)y^2 & d \equiv 1 \pmod 4 \end{cases}$

$$f(x,y) = ax^2 + bxy + cy^2$$

then $4af(x,y) = (2ax+by)^2 - dy^2$

$d > 0$: f is positive (or negative) definite. i.e always same sign (or zero)

$d < 0$: f is indefinite. i.e takes both signs

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equivalence of two quad forms

$X = \begin{pmatrix} x \\ y \end{pmatrix}$

$X' = \begin{pmatrix} x' \\ y' \end{pmatrix}$

① quad form

$F = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}$

$f = X^T F X$

② integral unimodula substitution

$U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ $\begin{cases} \det U = 1 \\ p, q, r, s : \text{integers} \end{cases}$

(i) $X = UX'$

(ii) $F' = U^T F U$ (equivalent)

- reflexive (I), symmetric (U^{-1}), transitive (U_1U_2)
- values assumed by equivalent forms

as (x,y) runs through ~~integers~~ relatively prime integers are same.

Note.

$U^{-1} = \begin{pmatrix} s & -q \\ -r & p \end{pmatrix}$

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equivalent forms have same discriminant

$\because$ $\det F = ac - \frac{b^2}{4}$

$\det F' = \det U^T F U = 1 \cdot (ac - \frac{b^2}{4}) \cdot 1$

Properties of integral unimodula substitution

let $U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ with $\begin{cases} \det U = 1 \\ p, q, r, s : \text{integers} \end{cases}$

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① U is bijection $(\mathbb{Z}^2 \longrightarrow \mathbb{Z}^2)$

(i) injection : $UX_1 = UX_2$ then $U^{-1}UX_1 = U^{-1}UX_2$

$X_1 = X_2$

(ii) surjection : $Y \in \mathbb{Z}^2$ then $X = U^{-1}Y \in \mathbb{Z}^2$

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② $X = UX'$ then

$$\begin{aligned} f'(X') &= X'^T F' X' \\ &= X'^T U^T F U X' \\ &= X^T F X = f(X) \end{aligned}$$

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③ $(x, y) = 1$ iff $(x', y') = 1$

$\because$ $x = px' + qy'$ then (x', y') divides both x, y

$y = rx' + sy'$ however $(x, y) = 1$

2. Reduction

positive definite quad form $\Rightarrow$ $\underbrace{d<0}_{\text{definite}}$ and $\underbrace{a>0}_{\text{positive}}$ $\Rightarrow$ $c>0$

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transform f into binary form s.t $|b| \le a \le c$

$F = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}$

① replace $a>c$ by $a<c$ $\quad U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ then $F' = \begin{pmatrix} c & -\frac{b}{2} \\ -\frac{b}{2} & a \end{pmatrix}$

② replace $|b|>a$ by $|b| \le a$ $\quad U = \begin{pmatrix} 1 & \pm 1 \\ 0 & 1 \end{pmatrix}$ then $F' = \begin{pmatrix} a & \frac{b \pm 2a}{2} \\ \frac{b \pm 2a}{2} & a \pm b + c \end{pmatrix}$

- process terminates $\because$ a decreasing!
- reduced forms

(i) $-a < b \le a < c$ $\quad$ $\because$ 최신 ② $\Rightarrow$ $F' = \begin{pmatrix} a & \frac{a}{2} \\ \frac{a}{2} & c \end{pmatrix}$

i.e $-a < b$

(ii) $0 \le b \le a = c$ $\quad$ $\because$ 최신 ① $\Rightarrow$ $F' = \begin{pmatrix} c & -\frac{b}{2} \\ -\frac{b}{2} & a \end{pmatrix}$

i.e if $a=c$ then $0 \le b$

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class number $h(d)$ $=$ # of reduced forms with discriminant d

$=$ # of inequivalent classes of binary quad forms with discriminant d

i.e any two reduced forms are not equivalent.

$h(d)$ is finite.

$\because$ $-d = 4ac - b^2 \ge 3ac$

then $|b|, a, c \le |d|/3$

e.g $d=-4$ $\Rightarrow$ $3ac \le 4$

$a=1,\ c=1$

$\Rightarrow$ $b=0$

$\Rightarrow$ $h(-4)=1$

Claim if $\underbrace{\text{reduced}}_{\substack{-a<b\le a\le c \\ 0\le b\le a=c}}$ form f, f' are $\underbrace{\text{equivalent}}_{F'=U^TFU}$ then $\begin{cases} a'=a \\ b'=\pm b \\ c'=c \end{cases}$

- if $|x|\ge|y|$ (x, y : non-zero integers)

then $f(x,y) \ge |x|(a|x|-|by|)+c|y|^2$

$\ge |x|^2(a-|b|)+c|y|^2$

$\ge a-|b|+c$

- smallest values assumed by f, f' with $(x,y)=1$

$(\pm1, 0) \xrightarrow{f,f'} a$

$(0, \pm1) \longrightarrow c$

$(\pm1,\pm1)$ or $(\pm1,\mp1) \longrightarrow a-|b|+c$

$\Rightarrow a'=a,\ b'=\pm b,\ c'=c$

Claim if $b'=-b$ then $b=0$

- consider only $-a<b<a<c$

because (i) $-a<b$ (reduced f)

$-a<-b$ (reduced f')

(ii) $a=c$ then $b\ge 0$ (reduced f)

$-b\ge 0$ (reduced f')

- $f(x,y)\ge a-|b|+c>c>a$ (x, y : non-zero integers)

- $U=\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ then

$a=f'(\pm1,0)=f(p,r) \rightsquigarrow p=\pm1,\ r=0$

$c=f'(0,\pm1)=f(q,s) \rightsquigarrow q=0,\ \underbrace{s=\pm1}_{ps-qr=1}$

- $U=\begin{pmatrix} \pm1 & 0 \\ 0 & \pm1 \end{pmatrix}$ then $F'=U^TFU=F$

3. Representation by binary forms.

binary form f with discriminant d s.t. $n = f(p, r)$ with $(p, r) = 1$

$\Longleftrightarrow$ $x^2 \equiv d \pmod{4n}$ solvable

(n is properly represented by f)

$(\Rightarrow)$

- $U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ $\because$ $(p, r) = 1$, $ps - qr = 1$
- $n = f(p, r)$
  $= f'(1, 0) = a'$
- $b'^2 - 4nc' = d$

  $\because$ same discriminant

$(\Leftarrow)$

- $b^2 - \underbrace{4n}_{= a}\underbrace{c}_{= c} = d$

  $\because$ $b^2 \equiv d \pmod{4n}$
- $a = f(1, 0) = n$

$(n, d) = 1$ then # of proper representations of n by all reduced form with discriminant d $= \omega S$

① S : # of solutions of $x^2 \equiv d \pmod{4n}$ with $0 \le x < 2n$

② ω : # of $\underbrace{\text{automorphs}}_{U \text{ that takes } f \text{ into itself}}$ of a reduced form

c.f related to Pell equation.

$$\omega = \begin{cases} 2 & (d < -4) \\ 4 & (d = -4) \\ 6 & (d = -3) \end{cases}$$

$\rightarrow$ $\begin{cases} x = x' \\ y = y' \end{cases}$ or $\begin{cases} x = -x' \\ y = -y' \end{cases}$

4. Sums of two squares.